JGH Open An open access journal of gastroenterology and hepatology
Open Access
JGHF Journal of Gastroenterology and Hepatology Foundation
doi:10.1002/jgh3.12885

CASE REPORT

Collagenous colitis following SARS-Cov2 mRNA vaccination

Momoko Iketani, Yutaka Takada and Toshinao Itani

Department of Gastroenterology and Hepatology, Kobe City Nishi-Kobe Medical Center, Kobe, Japan

Key words
collagenous colitis, microscopic colitis, SARS-Cov2 mRNA vaccination.

Accepted for publication 16 February 2023.

Correspondence
Yutaka Takada, Department of Gastroenterology and Hepatology, Kobe City Nishi-Kobe Medical Center, 5-7-1, Kojidai, Nishi-ku, Kobe-shi, 651-2273, Japan.
Email: ichigojellylike@gmail.com

Declaration of conflict of interest: The authors have no conflicts of interest to disclose concerning this article.

Abstract

A healthy 49-year-old female developed intractable watery diarrhea after the Pfizer SARS-Cov2 mRNA vaccination. She was subsequently diagnosed with collagenous colitis (CC). She had no prior history of medication use, suggesting of vaccination being the trigger. CC or lymphocytic colitis should be considered as differential diagnoses for persistent watery diarrhea after SARS-Cov2 mRNA vaccination.

Introduction

Collagenous colitis (CC) and lymphocytic colitis (LC) are inflammatory bowel diseases characterized by chronic, watery, non-bloody diarrhea. Several drugs and other autoimmune diseases have been associated with CC and LC. In addition to this, these days, a few cases of CC or LC have been reported as adverse events of SARS-Cov2 mRNA vaccination. Here we report a case of CC occurring after the Pfizer SARS-Cov2 mRNA vaccination.

Case report

A 49-year-old otherwise healthy woman presented with intractable watery diarrhea (10–20 times/day) without abdominal pain or

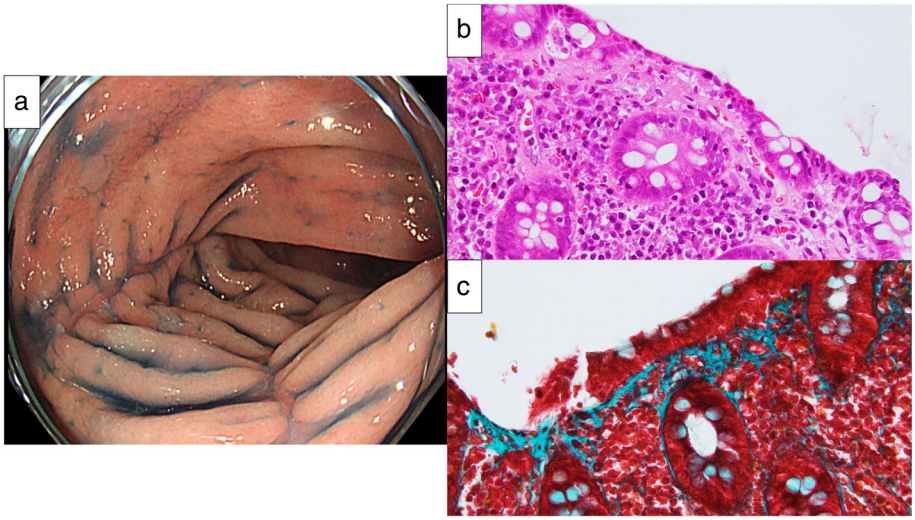


Figure 1 (a) Colonoscopy showed ulcer scars running longitudinally across the sigmoid colon. (b) Pathological examination (H&E) showed infiltration of lymphocytes and plasma cells into the lamia propria, and a subepithelial collagen band over 10 μm. (c) Masson's trichrome staining showed the subepithelial collagen band more clearly.

© 2023 The Authors. JGH Open published by Journal of Gastroenterology and Hepatology Foundation and John Wiley & Sons Australia, Ltd.
This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

bloody stools. The symptoms began 3 days after her first Pfizer SARS-Cov2 mRNA vaccination, but improved after 2 months. However, 3 days after her third vaccination, watery diarrhea flared up; antidiarrheal medications did not work. Despite frequent watery diarrhea, the patient's general condition was relatively good, with no abdominal pain, fever, arthritis, or skin rash. A stool culture test was negative, and a blood test showed no obvious abnormalities including HbA1c, thyroid function, IgA, IgG, IgM, and antinuclear antibodies. Colonoscopy showed ulcer scars running longitudinally across the sigmoid colon (Fig. 1a), and CC was diagnosed by biopsy, which showed a thickened subepithelial collagen band (Fig. 1b,c). Ninety days later, the diarrhea had spontaneously resolved.

Discussion

CC is a subtype of microscopic colitis (MC), in addition to LC.[1] CC is often caused by drugs such as antiacids and nonsteroidal anti-inflammatories.[2] An association with inflammatory or autoimmune diseases such as rheumatoid arthritis, thyroid disorders, coeliac disease, and diabetes mellitus have also been suggested.[3] A characteristic endoscopic finding of CC is longitudinal ulcers called mucosal tears. The longitudinal ulcers with well-defined borders and lack of surrounding edema and erythema are different from those of ulcerative colitis and Crohn's disease. The definitive diagnosis is based on histopathological examination, which shows thickening of the collagen band just below the epithelium to more than 10 μm and inflammatory cell infiltration in the submucosa.[4] For management, patients should be advised to avoid medications associated with MC and use antidiarrheals. In active cases, oral budesonide may be effective for achieving induction and maintenance of both clinical and histological response.[5] Our case had no prior history of medication use other than antidiarrheals, and there were no symptoms or laboratory findings suspicious of other inflammatory or autoimmune diseases. Symptoms occurred a few days after vaccination and resolved within 2 months, suggesting of vaccination being the trigger. The Centers for Disease Control and Prevention's vaccine adverse event database shows four cases of CC and five cases of LC following SARS-Cov2 mRNA vaccination. Therefore, CC or LC should be considered as differential diagnoses for persistent watery diarrhea after SARS-Cov2 mRNA vaccination.

Patient consent statement. Informed consent was obtained from the patient for publication of this case report and accompanying images.

References

1 Nguyen GC, Smalley WE, Vege SS *et al*. American Gastroenterological Association Institute Guideline on the medical management of microscopic colitis. *Gastroenterology*. 2016; **150**: 242–6.
2 Beaugerie L, Pardi DS *et al*. Review article: drug-induced microscopic colitis – proposal for a scoring system and review of the literature. *Aliment. Pharmacol. Ther.* 2022; **22**: 277–84.
3 Bohr J, Tysk C, Eriksson S, Abrahamsson H, Jarnerot G. Collagenous colitis: a retrospective study of clinical presentation and treatment in 163 patients. *Gut*. 1996; **39**: 846–51.
4 Liszka L, Woszczyk D, Pajak J. Histopathological diagnosis of microscopic colitis. *J. Gastroenterol. Hepatol.* 2006; **21**: 792–7.
5 Sebastian S, Wilhelm A, Jessica L, Myers S, Veysey M. Budesonide treatment for microscopic colitis: systematic review and meta-analysis. *Eur. J. Gastroenterol. Hepatol.* 2019; **31**: 919–27.

© 2023 The Authors. JGH Open published by Journal of Gastroenterology and Hepatology Foundation and John Wiley & Sons Australia, Ltd.